How it works

The circuit has two main units: the control unit and the relay/appliance switching unit. The circuit diagram is shown in Fig. 3.

EFY note. Fig. 3 shows only two appliance channels (APPL1 and APPL16). The remaining channels, APPL2 to APPL15, with their relays (RL2-RL15) and transistors (T2-T15), are identical and are not shown.

Control unit. This part is built around IC1 (CD4093), IC2 (CD4029) and IC3 (74LS154). Switches S1 (increment) and S2 (decrement) are used to select the required appliance. Resistors R2 and R3 set the required logic levels at the inputs. When either push-to-on switch is pressed, its signal goes to the Schmitt-trigger NAND gates IC1A and IC1B.

The Schmitt-trigger gates remove the unwanted transitions caused by mechanical contact bounce, so each press produces one clean logic pulse instead of several false ones. The remaining gates, IC1C and IC1D, together with C1 and R4, provide a continuous pulse and make sure the counter triggers reliably.

The clean pulses go to IC2 (CD4029), which is configured as an up/down counter. Pressing S1 increases the count by one, and pressing S2 decreases it by one. The four counter outputs are the selected channel in binary form. Four bits represent 16 states (S0 to S15), so up to 16 channels can be selected.

The 4-bit count is fed to the address inputs (A0 to A3) of IC3, the 74LS154 decoder. The decoder converts the count into one of 16 individual outputs, AP1 to AP16,

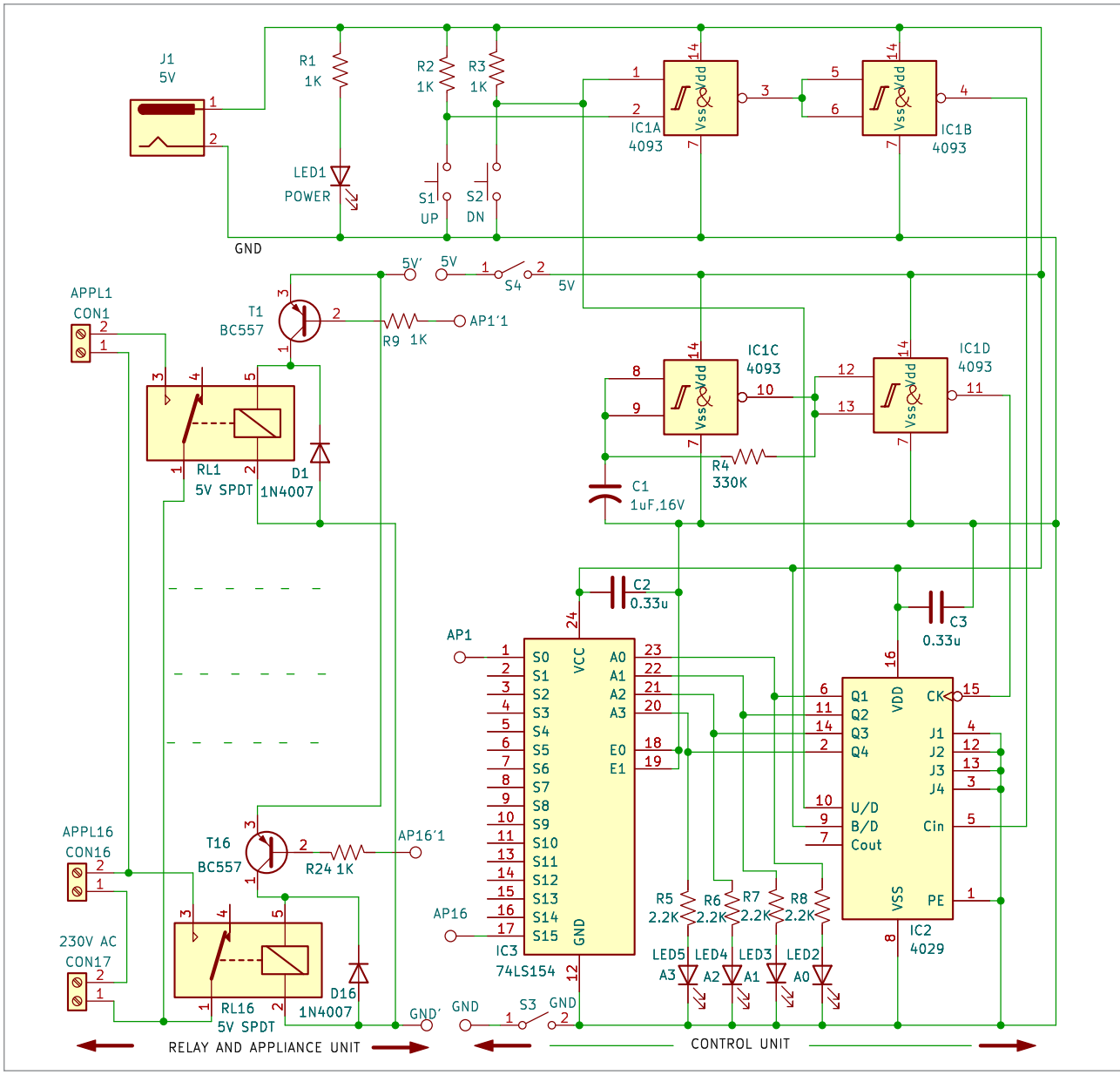


Fig. 3: Circuit diagram